SourceLoop

AI agents that source parts for a company, with people deciding the big calls and software enforcing the rules. Proof of concept for SAP Business Network (synthetic data).

The problem

Buying a part from a new supplier is slow and manual: finding suppliers, checking certificates and sanctions, chasing quotes, negotiating, drafting the contract. One purchase can take weeks and dozens of hand-offs, and a rushed check can let a risky supplier through.

How it works

1 Need	2 Discovery	3 Qualification	4 Sourcing	5 Contracting	6 Onboarding
Buyer says what they need	Agent shortlists suppliers	Certificates, sanctions, risk	Quotes and negotiation	Clauses, signature	Hand-off to operations

Who	Does what
Buyer agent and **supplier agent**	Each works for one company. The buyer agent finds and screens suppliers and negotiates. Each supplier has its own agent that quotes within its own limits. Neither side ever sees the other's limits.
Neutral referee (plain software, not AI)	Passes messages between the two sides, enforces price limits and rules in code, reviews every negotiation turn for fair conduct, and writes every step to a tamper-evident record.
People	Make the big calls: compliance flags, technical pass or fail, the award, protected contract clauses (liability, termination, IP), signature, bank details, releasing payment. The agent recommends; a person decides.

Why a business can trust it

- **Limits live in code, not in the AI.** A supplier agent cannot go below its floor and a buyer agent cannot exceed its budget, however it is prompted.
- **Sanctions and certificate checks are rules,** never the AI's opinion, and a sanctions hit is never cleared automatically.
- **Everything is on the record.** Each step is chained to the one before, so any later change is detectable.
- **Attacks are contained.** A hostile news article, inflated supplier claims and attempts to break limits are tested scenarios.

Early results (simulated, 12 sample needs)

	With agents	Manual process (modelled)
Time to a deal, working days (mean)	3.8	45.9
Human touches per purchase (mean)	6	41
Risky suppliers reaching the shortlist	0	3
Purchases reaching a deal	11 of 12	12 of 12 (to award)

The agent figures come from real runs on synthetic data, but the time they save is simulated and the manual figures are an assumption, not a measured study. Read them as direction, not proof.

Demo in 5 minutes

1. Open **http://localhost:3010** and choose **Buyer** (no password). Click **Run a sample project**.
2. Watch the six stages advance. The card at the top always says what is happening and what is needed next.
3. When it says **Needs a decision**, click **Review and decide** and approve. Then sign out and pick **Compliance**, **Quality**, **Legal** and **Finance** in turn: each approves their own checkpoint.
4. Pick **Supplier** to show the other side: it sees only its own conversation, never the buyer's limits.
5. Open the finished project: suppliers screened, price chart, negotiation review and the verified audit record.

What this is and is not

A working test bed on made-up data. The links to SAP and commercial data sources are stubs, the default demo replays recorded agent answers (a live mode uses a small local AI model), and there is no production login or integration yet. Next: real data connectors, a one-login demo, and outcome dashboards for buyers.